Case study

1. Setup

Every controller runs in every situation 100 times, with the seeds 1 to 100. All controllers use the same seeds, so they have the same random numbers. A run has 400 steps. The six situations are: S1 is the start of the program, S2 is an unknown inflow change (u_0 goes from 50 to 62), S3 is a set point change (from 74 to 20), S4 is heavy rain and then strong evaporation, S5 is S2 with noise on the level reading, and S6 is an extreme change of the flows (u_0 goes from 50 to 75).

The weights of the controllers come from the tuning done in modeling of the experimental part:

Controller	Weights
P	K_P= 6
PI	K_P= 3, K_i= 0.2
PID	K_P= 14, K_i= 0.02, K_d= 100
PID+	K_P= 4, K_i= 0.2, K_d= 10
Ours	K_P= 10, K_i= 0.2, K_d= 25, k = 0.1 for f_p, 0.01 for f_i, 0 for f_d, dead-band 0.2

We use the metrics from chapter 6 of the modeling of the experimental part. A smaller IAE means a smaller total error.

2. E1: comparison with P, PI, PID and PID+

Table 4.1 shows the mean IAE of 100 runs, with the standard deviation in brackets.

IAE mean (std)	S1	S2	S3	S4	S5	S6
P	1444 (124)	2450 (125)	2244 (142)	2014 (281)	2474 (118)	3665 (144)
PI	1843 (76)	1963 (76)	2768 (102)	2476 (230)	2015 (76)	2272 (167)
PID	1441 (76)	1394 (90)	1897 (90)	1942 (234)	1633 (107)	1919 (110)
PID+	1142 (147)	1262 (147)	1776 (180)	1733 (193)	1379 (139)	1588 (165)
Ours	1083 (104)	1270 (102)	1631 (130)	1560 (206)	1606 (98)	1757 (118)

Table 4.1. The IAE of the five controllers in the six situations (E1)

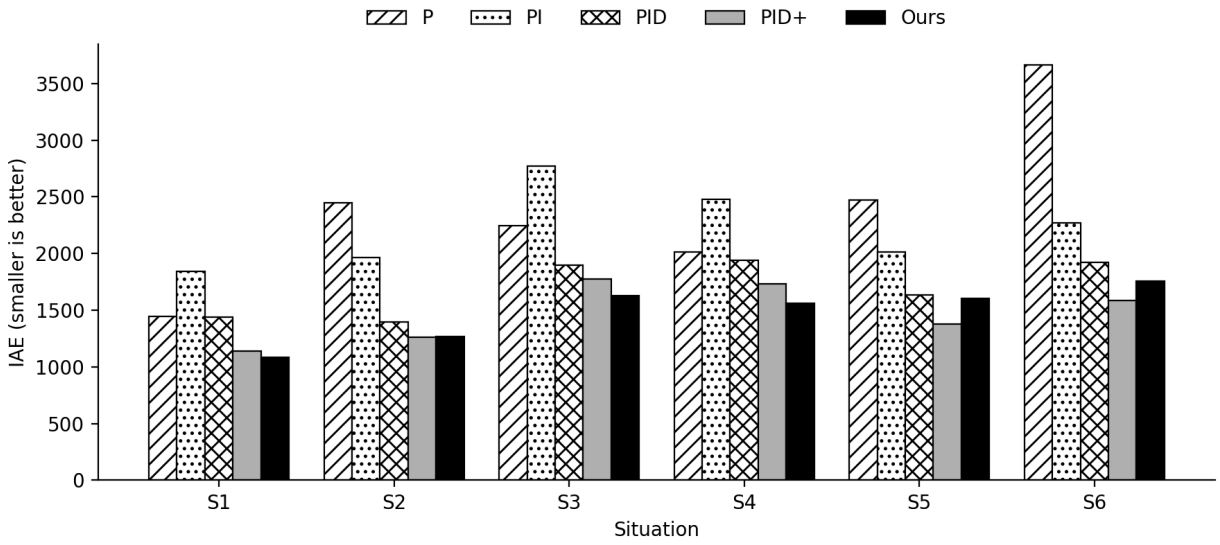


Figure 4.1. The IAE of the five controllers in the six situations (E1)

Table 4.2 shows in how many runs (percent) the level stays under 1 percent from the set point until the end, and Table 4.3 shows the valve movement.

Settled runs (%)	S1	S2	S3	S4	S5	S6
P	100	0	100	100	0	0
PI	100	100	100	100	100	100
PID	63	73	82	56	41	17
PID+	100	100	100	100	96	100
Ours	100	81	100	100	22	75

Table 4.2. The percent of settled runs (E1)

Valve movement	S1	S2	S3	S4	S5	S6
P	164	188	303	439	772	254
PI	119	139	221	328	479	203
PID	1130	1127	1131	1138	1158	1131
PID+	100	136	191	390	900	180
Ours	154	258	278	533	1047	275

Table 4.3. The valve movement (the sum of the changes of the valve position) (E1)

Because the runs are paired, we also count in how many runs our controller has a smaller IAE than the other one (Table 4.4). The last column is the mean difference of the IAE between PID+ and ours, so a negative number means that PID+ is better.

Ours is better in (of 100 runs)	vs P	vs PI	vs PID	vs PID+	Mean IAE difference vs PID+
S1	100	100	100	63	59
S2	100	100	91	35	−8
S3	100	100	95	86	145
S4	100	100	96	87	173
S5	100	100	63	6	−227
S6	100	100	92	15	−169

Table 4.4. The paired comparison of the IAE (E1)

What we see:

- **P:** the level stays away from the set point when u_0 is not 50. The final error is 4.00 in S2 and S5 and 8.33 in S6. This is what we calculated in 3.3: with K_P = 6 the valve is u^* = 50 − 3 * e, so for u_0 = 62 we get e = −4, and for u_0 = 75 we get e = −8.33. P never settles in S2, S5 and S6.
- **PI:** the integral removes this error, and PI settles in all the runs. But it is slow at the start and when the set point changes (the overshoot is 25.8 in S1 and 19.7 in S3), and it has the biggest IAE in S3 and S4.
- **PID:** it has about the same valve movement in all the situations (1127 to 1158), 7 times more than ours in S1. The reason is that the derivative of the raw error reacts to the random gain r(k) of the valve. It settles only in 17 to 82 percent of the runs.
- **PID+:** it is the strongest of the four. The filtered derivative and the anti-windup fix the problems of the PID.
- **Ours against P, PI and PID:** our IAE is smaller in all the 100 runs against P and PI in every situation, and in 91 to 100 runs against PID (63 in S5).
- **Ours against PID+:** ours is better in S1 (63 runs of 100), S3 (86) and S4 (87). In S2 the two are almost equal (ours is better in 35 runs, the mean difference is −8). In S5 and S6 PID+ is clearly better (ours is better in only 6 and 15 runs).
- **Start and set point change:** in S1 and S3 ours is the fastest (46 and 30 steps to settle, against 57 and 49 for PID+) and it has a smaller overshoot than PID+ (8.6 against 13.0 in S1, 5.9 against 10.3 in S3). The valve of PID+ changes direction 198 times in S1, and ours only 28 times (in S3, 257 against 48; in S4, 234 against 82).
- **Valve movement:** ours moves the valve more than PID+ in all the situations (154 against 100 in S1, 533 against 390 in S4).

Figure 4.2 shows one run of S3 (seed 1). After the set point goes down to 20, ours reaches it faster than PID+.

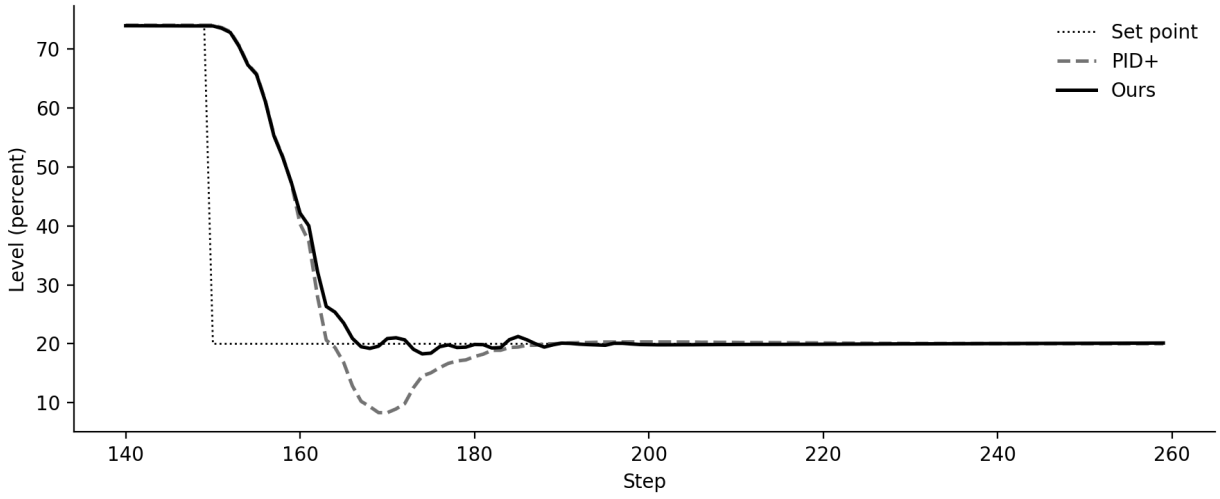


Figure 4.2. The level in S3 for PID+ and for our controller (seed 1)

E2, E3 and E4 in the same results

- **E2, noise (S5):** all the controllers move the valve much more with the noise on the reading. Ours moves it 1047 (4 times more than in S2), and PID+ moves it 900 (6.6 times more than in S2). Ours settles in only 22 percent of the runs, and PID+ in 96 percent. The noise (0.5) is bigger than our dead-band (0.2), so the dead-band does not protect the valve, and the derivative is cancelled 17 times in a run.
- **E3, sudden change of the inflow (S2 and S6):** this is the case where our controller is weaker. In S2 the integral has to keep the valve at 62 and in S6 at 75. In chapter 3 of the modeling of the experimental part we said that the integral reset clears the accumulated error when the error changes sign. Figure 4.3 shows S2 for one run: ours comes back faster at the start, but then it needs a long time to get to the set point, and PID+ is on the set point from step 230.
- **E4, rain and evaporation (S4):** ours has the smallest IAE (1560) and settles in all the runs. The maximum error after the event is almost the same for all (from 18.4 to 20.3), because the rain adds water faster than the valve can take it out. So the controllers differ in how they recover, not in the maximum error.

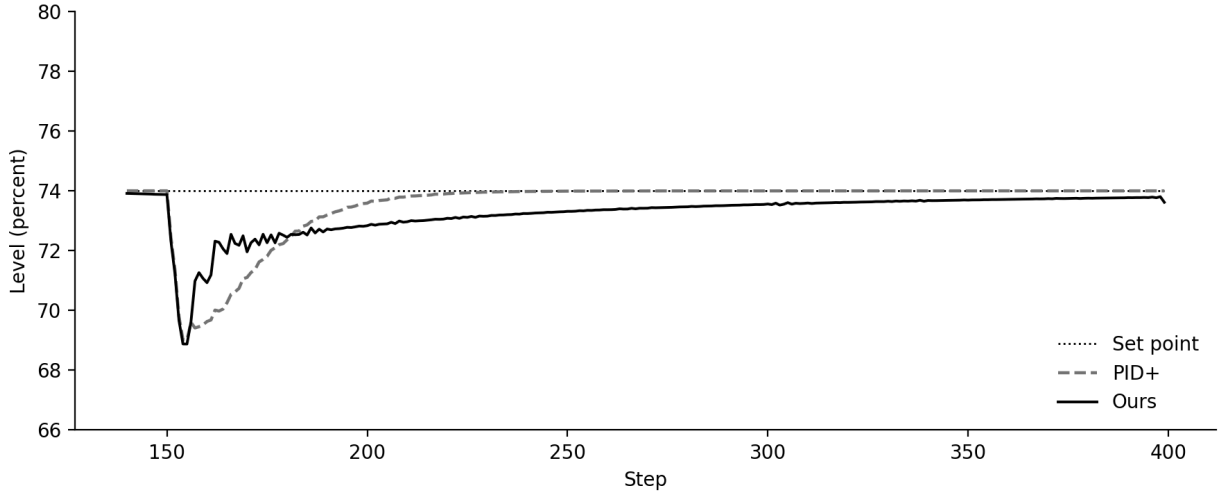


Figure 4.3. The level in S2 for PID+ and for our controller (seed 1)

3. E5: the cancel of the derivative

In E1 the derivative is cancelled 2.4 times in a run in S1, 3.5 in S2, 4.9 in S3, 11.0 in S4, 17.0 in S5 and 4.3 in S6. To see if it helps, we turn it off (E6). The IAE without the cancel is 1086, 1273, 1644, 1557, 1609 and 1760, against 1083, 1270, 1631, 1560, 1606 and 1757 with the cancel. The difference is at most 13 (less than 1 percent), so we do not see an effect. We did not test a case where the level really runs away, because even in S6, the biggest change, the maximum error is 26 and the level comes back.

4. E6: turning off one part at a time

IAE mean	S1	S2	S3	S4	S5	S6
Full controller	1083	1270	1631	1560	1606	1757
No second derivative	1133	1337	1720	1689	1710	1844
No smoothing	1432	1737	2262	1934	1996	2213
No dead-band	1052	1257	1601	1530	1620	1750
No integral reset	1927	1715	2411	2279	1812	2093
No cancel of the derivative	1086	1273	1644	1557	1609	1760
Linear fp	1371	1559	2099	1900	1864	2149
Linear fi	1453	1686	2100	1932	1954	2207
Linear fp and fi	1642	1863	2484	2171	2140	2408

Table 4.5. The IAE when one part is turned off (E6)

Settled runs (%)	S1	S2	S3	S4	S5	S6
Full controller	100	81	100	100	22	75
No second derivative	100	55	100	100	18	85
No smoothing	98	34	96	96	33	28
No dead-band	100	98	100	100	17	75
No integral reset	100	100	100	100	70	99
No cancel of the derivative	100	77	100	100	17	76
Linear f_p	100	86	100	100	24	15
Linear f_i	100	48	100	100	32	27
Linear f_p and f_i	100	64	100	100	19	17

Table 4.6. The percent of settled runs when one part is turned off (E6)

- **Smoothing:** it is the most important part. Without it the IAE is bigger by 24 to 39 percent, and the valve movement is 2 to 4 times bigger in most situations (665 against 154 in S1, 1051 against 258 in S2, 877 against 278 in S3).
- **Nonlinear functions fp and fi:** with the line f(x) = x instead of f_p, the IAE is bigger by 16 to 29 percent. With the line instead of f_i it is bigger by 22 to 34 percent, and with the line in both by 33 to 52 percent. In S1 the overshoot grows from 8.6 to 23.1 (f_p), 25.1 (f_i) and 25.6 (both), and in S3 from 5.9 to 18.0, 14.1 and 19.6. We tuned the weights for the nonlinear functions and we did not tune them again for the line, so this comparison helps the nonlinear functions. Figure 4.5 shows the functions.
- **Integral reset:** without it the IAE is much bigger in S1 (1927 against 1083), S3 (2411 against 1631) and S4 (2279 against 1560), and the settling in S1 takes 298 steps instead of 46. But when u_0 is not 50 the reset is a problem: without it the runs settle more often (S2 100 against 81 percent, S5 70 against 22, S6 99 against 75) and the final error is smaller (S5 0.71 against 1.78). The overshoot does not get smaller with the reset (S1: 9.7 without it and 8.6 with it; S3: 5.2 without it and 5.9 with it).
- **Second derivative:** without it the IAE is bigger by 5 to 8 percent and the settling is slower (S1: 60 steps against 46, S3: 51 against 30). But the overshoot is smaller (S1: 2.7 against 8.6, S3: 2.4 against 5.9). So it makes the controller faster, with more overshoot.
- **Dead-band:** without it the IAE is a bit smaller in five situations (for example 1052 against 1083 in S1), and the runs settle more often in S2 (98 against 81 percent). But the valve changes direction

2 to 3 times more often (S1: 73 against 28, S3: 143 against 48, S4: 192 against 82). So the dead-band protects the valve and it costs a little accuracy.

- **Cancel of the derivative:** see 4.3, there is no effect.

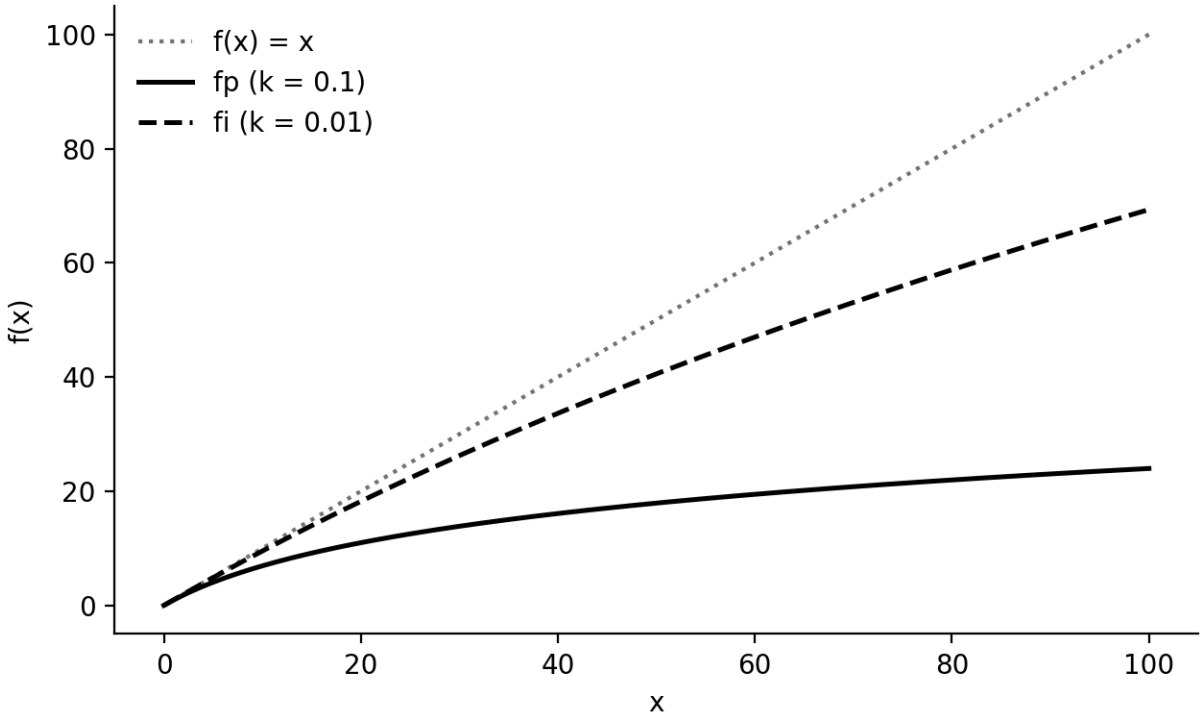


Figure 4.5. The functions f_p(k = 0.1) and f_i(k = 0.01) compared with the line f(x) = x

5. E7: changing the weights

We change one weight at a time, in S2, S3 and S4. The numbers in brackets show how much the IAE changes against the original weights (K_p = 10, K_i = 0.2, K_d = 25).

IAE (change)	S2	S3	S4
Original weights	1270	1631	1560
K_p × 0.5	1524 (+20%)	1803 (+11%)	1734 (+11%)
K_p × 0.8	1302 (+3%)	1644 (+1%)	1561 (0%)
K_p × 1.25	1327 (+4%)	1724 (+6%)	1641 (+5%)
K_p × 2	1773 (+40%)	2427 (+49%)	2293 (+47%)
K_i × 0.5	1417 (+12%)	1611 (−1%)	1539 (−1%)

IAE (change)	S2	S3	S4
K_i × 0.8	1304 (+3%)	1610 (−1%)	1535 (−2%)
K_i × 1.25	1331 (+5%)	1665 (+2%)	1588 (+2%)
K_i × 2	1442 (+14%)	1829 (+12%)	1700 (+9%)
K_d × 0.5	1591 (+25%)	2149 (+32%)	1957 (+25%)
K_d × 0.8	1315 (+4%)	1707 (+5%)	1609 (+3%)
K_d × 1.25	1339 (+5%)	1657 (+2%)	1569 (+1%)
K_d × 2	1583 (+25%)	1916 (+17%)	1808 (+16%)

Table 4.7. The IAE when one weight is changed (E7)

A change of 20 to 25 percent changes the IAE with at most 6 percent in almost all the cases (the exception is K_i × 0.5 in S2, with +12 percent). Halving or doubling a weight makes the result much worse (up to +49 percent). So the controller does not need a perfect tuning, but it needs weights in the right area. In S3 and S4 a smaller K_i is a little better (−1 to −2 percent), so the weights from the tuning are good, but not the best possible. The settled runs in S2 are sensitive to the weights (for example 44 percent for Kp × 0.8 and 100 percent for Kd × 0.5), because S2 is the situation where the integral reset hurts.

6. E8: the manual test

In the program we stop the process with the box Process runs, and the controller goes on. We start with level 10 and the set point 74. The controller waits 5 steps, then the valve goes up with 3 percent per step and it reaches 100 at step 21, and the integral goes down to its limit (−1500) at step 23. Both stay there, because the level does not change. So the controller does not stop after one proportional step. The integral keeps adding the error, and the valve moves in the direction opposite to the error (the level is under the set point, so the valve opens). This is what we described in chapter 3 of the modeling and it is the reason why the integral limit is needed.

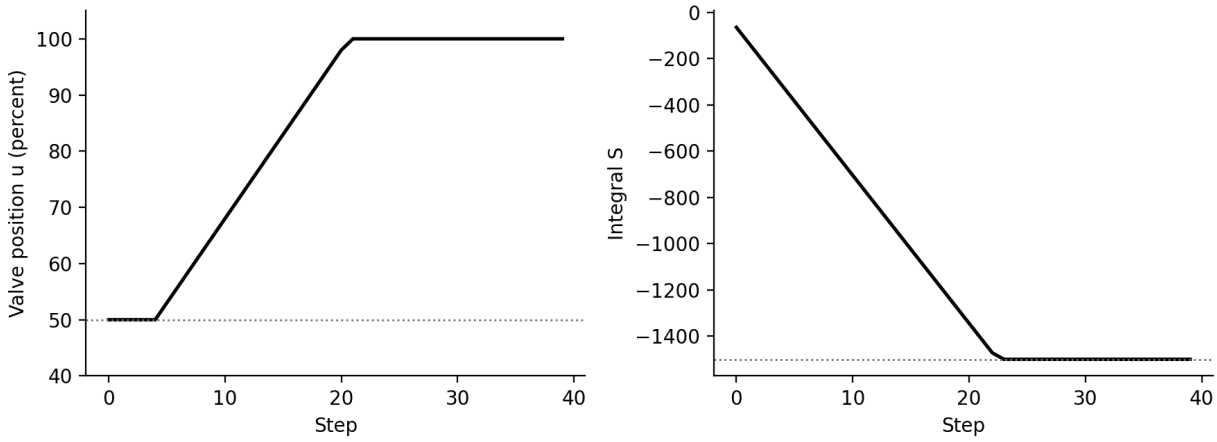


Figure 4.4. The valve and the integral when the process is stopped (E8)

7. Answers to the research questions

- **RQ1, does the second derivative help the controller react earlier?** Partly. It makes the settling faster (S1: 46 steps against 60, S3: 30 against 51) and the IAE smaller by 5 to 8 percent, but the overshoot is bigger (8.6 against 2.7 in S1, 5.9 against 2.4 in S3).
- **RQ2, are f_p and f_i better than f(x) = x?** Yes in our tests. With the line the IAE is bigger by 16 to 52 percent in all the situations, and the overshoot in S1 is about 3 times bigger. But the weights were tuned for the nonlinear functions.
- **RQ3, does the integral reset reduce the overshoot?** No, the overshoot is almost the same. The reset reduces the total error by 46 to 78 percent in S1, S3 and S4 and the settling time in S1 from 298 to 46 steps. But when u_0 is not 50 it stops the level from settling (S2: 81 percent against 100, S5: 22 against 70, S6: 75 against 99).
- **RQ4, does the cancel of the derivative stop the divergence?** We did not find an effect. The IAE changes with less than 1 percent, and we did not have a run where the level really diverges.
- **RQ5, how does the whole controller compare with P, PI, PID and PID+?** It is better than P, PI and PID in all the six situations (5 of the 6 against PID in more than 90 percent of the runs). Against PID+ it is better in S1, S3 and S4, almost equal in S2, and worse in S5 and S6.

8. Discussion

Our hypothesis was that the modified PID keeps the level closer to the set point and moves the valve less than a normal PID, when the flows are unknown. The results say:

- Closer to the set point than P, PI and a plain PID: yes, in all six situations.
- Less valve movement than a plain PID: yes (154 to 1047 against 1127 to 1158). Less than PID+: no, PID+ moves the valve less in all the situations.
- Closer to the set point than PID+: only in S1, S3 and S4, that is when the set point changes or when there is rain and evaporation. When the null position of the valve changes (S2, S5, S6), PID+ is as good or better.

The condition that we wrote in chapter 8 of modeling (a smaller IAE than all the other controllers in S2, S4 and S6) is not met, because PID+ is better in S6 and equal in S2. So we do not say that our controller is better than all. We say that the nonlinear functions, the smoothing and the second derivative help it at the start and at a change of the set point, and that its weak point is the integral reset when the flows are unknown and when there is noise.

Two ideas for a future version, that we did not test: to reset the integral only when the error is big, and to use a dead-band bigger than the noise of the sensor.

Limits

- It is only a simulation, and the values come from 100 runs for each case (the standard deviation of the IAE is between 76 and 281).
- The weights of all the controllers were found with the same search and the same cost, but the search can stop at a local minimum. The variants with the line f(x) = x were not tuned again.
- In E7 we changed only one weight at a time.